

交互题

人员

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上周作业检查

上周作业链接: <https://cppoj.kids123code.com/contest/4952>

王向东老师周日十点半C++离散化								
刷新								
#	用户名	姓名	编程分	时间	A	B	C	D
1	zhouziyi	周子一	400	10991	100	100	100	100
2	yangjinsuo	杨瑾硕	300	7420	100	100	100	
3	yuzijia1	于子珈	300	9173	100	100	100	
4	qinxiansen	秦显森	300	9867	100	100	100	
5	siyunxin	司云心	200	8192	100	100		
6	xieyakai	谢亚楷	200	9507	100	100	0	
7	chenluoran	陈洛冉	200	10884	100	100	0	
8	lizihan	李子瀚	200	11332	100	100		
9	niutongze	牛同泽	200	12231	100	100		

本周作业

<https://cppoj.kids123code.com/contest/5144> (课上讲了 A ~ D 题, 课后作业是 D 题必做)

课堂表现

今天课上讲了 交互题 这个知识点, 今天的 D 题是今年省选的真题, 这道题比较难, 不要求同学们必须掌握, 能把前 3 道题掌握即可。

课堂内容

猜数 (IO交互版)

交互题模板

```
#include <bits/stdc++.h>

using namespace std;

int main()
{
    int l = 1, r = 1e9;
    while (true) {
        int mid = (l + r) / 2;
        cout << mid << endl;
```

```

    int x; cin >> x;
    if (x == 0) break;
    if (x == -1) l = mid+1;
    else r = mid-1;
}
return 0;
}

```

nextmin

题目允许错一个, 把所有位置问一遍即可

```

#include <bits/stdc++.h>

using namespace std;

const int maxn = 1000 + 5;
int w[maxn];

int main()
{
    int n; cin >> n;
    for (int i = 1; i <= n; ++i) {
        cout << "? " << i << " " << i << endl;
        int x; cin >> x;
        w[i] = x;
    }

    cout << "! ";
    for (int i = 1; i <= n; ++i) cout << w[i] << " ";
    return 0;
}

```

[COI 2020] Zagrade

括号匹配, 每次判断 i 和 栈顶 之间合不合法

```

#include <bits/stdc++.h>

using namespace std;

const int maxn = 1e5 + 5;
char s[maxn];

int main()
{
    int n, Q; cin >> n >> Q;
    stack<int> stk;
}

```

```

for (int i = 1; i <= n; ++i) {
    if (stk.empty()) stk.push(i);
    else {
        int j = stk.top();
        cout << "? " << j << " " << i << endl;

        int x; cin >> x;
        if (x == 1) s[j] = '(', s[i] = ')', stk.pop();
        else stk.push(i);
    }
}

int len = stk.size();
for (int i = 1; i <= len/2; ++i) {
    int x = stk.top(); s[x] = '('; stk.pop();
}
for (int i = 1; i <= len/2; ++i) {
    int x = stk.top(); s[x] = ')'; stk.pop();
}

cout << "! ";
for (int i = 1; i <= n; ++i) cout << s[i];
cout << endl;
return 0;
}

```

[省选联考 2026] 排列游戏 / perm

```

#include <vector>
void init(int, int);
std::vector<int> perm(int);
int query(int, int);

#include <bits/stdc++.h>

using namespace std;

const int maxn = 3e4 + 5;
int w[maxn];

void init(int c, int t) { memset(w, 0, sizeof(w)); }

struct node {
    int value, pos;
    bool operator < (const node& p) const { return value < p.value; }
    bool operator > (const node& p) const { return value > p.value; }
};

std::vector<int> perm(int n) {
    priority_queue<node, vector<node>, less<node>> q;
    set<int> s;
}

```

```
for (int i = 0; i <= n-1; ++i) s.insert(i);

int pos_0 = 0;
for (int i = n-2, last = n; i >= 0; i--) {
    int t = query(0, i);
    if (t < last) w[i+1] = t, last = t, s.erase(t);
    else q.push({last, i+1});

    if (t == 0) { pos_0 = i+1; break; }
}

for (int i = 1, last = n; i <= pos_0; ++i) {
    int t = query(i, n-1);
    if (t < last) w[i-1] = t, last = t, s.erase(t);
    else q.push({last, i-1});
}

if (pos_0 == 0) w[0] = 0, s.erase(0);

while (!q.empty()) {
    node u = q.top(); q.pop();
    w[u.pos] = *s.rbegin();
    s.erase(*s.rbegin());
}

vector<int> vec;
for (int i = 0; i <= n-1; ++i) vec.push_back(w[i]);
return vec;
}
```